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$$\begin{aligned} & \lim_{x \rightarrow -\frac{\pi}{3}} \frac{2 \sin x + \sqrt{3}}{9x^2 - \pi^2} \\ &= \frac{2 \sin(-\frac{\pi}{3}) + \sqrt{3}}{9(-\frac{\pi}{3})^2 - \pi^2} \\ &= \frac{-\sqrt{3} + \sqrt{3}}{\pi^2 - \pi^2} = \frac{0}{0} \\ &\stackrel{H}{=} \lim_{x \rightarrow -\frac{\pi}{3}} \frac{2 \cos x}{18x} \\ &= \frac{2 \cos(-\frac{\pi}{3})}{18(-\frac{\pi}{3})} \\ &= \frac{2 \cdot \frac{1}{2}}{-6\pi} \\ &= -\frac{1}{6\pi} \end{aligned}$$

References